WORKED SOLUTION

STEP III 1991, Question 16

Probability and Statistics • Joint and Marginal Distributions

STEP III 1991, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q16

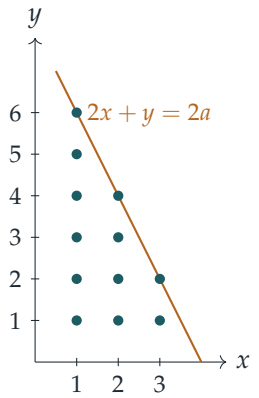
The key idea

Everything here is a finite sum over a triangle of lattice points. Draw the region first: for each x we need to know which y are allowed, and for each y which x are allowed. Once the limits are right, each marginal is the sum of an arithmetic progression. The only subtlety is that the largest allowed x for a given y is $\lfloor (2a-y)/2 \rfloor$, which depends on whether y is even or odd.

Solution

The region

We need $x \geq 1$, $y \geq 1$ and $2x + y \leq 2a$. Since $y \geq 1$, we need $2x \leq 2a - 1$, so $x \leq a - 1$. For a fixed x with $1 \leq x \leq a - 1$, the allowed values are $y = 1, 2, \ldots, 2a - 2x$. Likewise $y \leq 2a - 2$, because $x \geq 1$.



The figure shows the case $a = 4$: column x contains $2a - 2x$ points.

The marginal distribution of X and the value of c

Fix x and write $n = 2a - 2x$. Summing the arithmetic progression $2x+1, \ldots, 2x+n$,

$$\begin{aligned} \mathrm{P}(X = x) &= c \sum_{y=1}^{n} (2x + y) = c\left(2xn + \frac{n(n+1)}{2}\right) = c\,\frac{n}{2}\,(4x + n + 1) \\ &= c\,(a - x)(2a + 2x + 1), \qquad x = 1, 2, \ldots, a-1, \end{aligned}$$

and $\mathrm{P}(X = x) = 0$ otherwise.

The probabilities must add to 1. Put $k = a - x$, which runs from 1 to $a - 1$; then $2a + 2x + 1 = 4a + 1 - 2k$ and

$$\begin{aligned} \sum_{k=1}^{a-1} k(4a + 1 - 2k) &= (4a+1)\frac{(a-1)a}{2} - 2 \cdot \frac{(a-1)a(2a-1)}{6} \\ &= a(a-1)\left(\frac{4a+1}{2} - \frac{2a-1}{3}\right) = \frac{a(a-1)(8a+5)}{6}. \end{aligned}$$

Setting c times this equal to 1 gives

$$c = \frac{6}{a(a-1)(8a+5)},$$

as required.

The marginal distribution of Y

Fix y with $1 \le y \le 2a - 2$. The allowed x are $1, 2, \ldots, N$, where N is the largest integer with $2N \le 2a - y$. For N of these terms,

$$\sum_{x=1}^{N} (2x + y) = N(N+1) + yN = N(N + 1 + y).$$

y **even.** Then $2a - y$ is even and $N = \frac{1}{2}(2a - y)$, so $N + 1 + y = \frac{1}{2}(2a + 2 + y)$ and

$$\mathrm{P}(Y = y) = c \cdot \frac{(2a - y)(2a + 2 + y)}{4} = \frac{3(2a - y)(2a + 2 + y)}{2a(a-1)(8a+5)},$$

using the value of c. This is the required result.

? **Ask yourself** Why can't we use $N = \frac{1}{2}(2a - y)$ when y is odd?

When y is odd, $\frac{1}{2}(2a - y)$ is not an integer. The largest integer x with $2x \le 2a - y$ is $N = \frac{1}{2}(2a - y - 1)$: rounding down drops the half.

y **odd.** With $N = \frac{1}{2}(2a - 1 - y)$ we get $N + 1 + y = \frac{1}{2}(2a + 1 + y)$, so

$$\mathrm{P}(Y = y) = c \cdot \frac{(2a - 1 - y)(2a + 1 + y)}{4} = \frac{3(2a - 1 - y)(2a + 1 + y)}{2a(a-1)(8a+5)}.$$

(The two cases can be written as $3\big((2a+1)^2 - (y+1)^2\big)$ and $3\big(4a^2 - (y+1)^2\big)$ over the same denominator.)

The expectation of Y

Summing $y\,\mathrm{P}(Y=y)$ means handling even and odd y separately. It is cleaner to go back to the joint distribution and sum down each column. For fixed x, with $n=2a-2x$ as before,

$$\sum_{y=1}^{n} y(2x+y) = 2x\cdot\frac{n(n+1)}{2} + \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)}{6}\big(6x+2n+1\big).$$

Again put $k=a-x$, so $n=2k$ and $6x+2n+1=6a-2k+1$. The column sum becomes

$$\frac{2k(2k+1)(6a+1-2k)}{6} = \frac{k(2k+1)(6a+1-2k)}{3}.$$

Expanding, $k(2k+1)(6a+1-2k) = (6a+1)(2k^2+k) - 4k^3 - 2k^2$. Using the standard sums up to $a-1$,

$$\sum_{k=1}^{a-1}(2k^2+k) = \frac{a(a-1)(2a-1)}{3} + \frac{a(a-1)}{2} = \frac{a(a-1)(4a+1)}{6},$$

$$\sum_{k=1}^{a-1} k^3 = \frac{a^2(a-1)^2}{4}, \qquad \sum_{k=1}^{a-1} k^2 = \frac{a(a-1)(2a-1)}{6}.$$

So

$$\begin{aligned}
&\sum_{k=1}^{a-1} k(2k+1)(6a+1-2k) \\
&\quad= a(a-1)\left(\frac{(6a+1)(4a+1)}{6} - a(a-1) - \frac{2a-1}{3}\right) \\
&\quad= \frac{a(a-1)}{6}\big(24a^2+10a+1-6a^2+6a-4a+2\big) \\
&\quad= \frac{a(a-1)(6a^2+4a+1)}{2}.
\end{aligned}$$

Therefore

$$\begin{aligned}
\mathrm{E}(Y) &= \frac{c}{3}\cdot\frac{a(a-1)(6a^2+4a+1)}{2} = \frac{6}{a(a-1)(8a+5)}\cdot\frac{a(a-1)(6a^2+4a+1)}{6} \\
&= \frac{6a^2+4a+1}{8a+5}.
\end{aligned}$$

Checks

- For $a=2$ there are only two points, $(1,1)$ and $(1,2)$, with weights $3c$ and $4c$, so $c=\frac{1}{7}$. This agrees with $6/(2\cdot 1\cdot 21)$. Then $\mathrm{E}(Y)=\frac{3+8}{7}=\frac{11}{7}$, and the formula gives $\frac{33}{21}=\frac{11}{7}$.
- The marginal formulas for X and for Y (both parities), and $\mathrm{E}(Y)$, were checked by exact enumeration with fractions for every a from 2 to 29. For example, $a=3$ gives $\mathrm{E}(Y)=\frac{67}{29}$.

- For large a, $\mathrm{E}(Y) \approx \frac{3}{4}a$. The weight $2x + y$ favours points near the slanted edge, so Y sits well above the $\frac{2}{3}a$ you would get with equal weights on the triangle.

Common slips

- Letting x run up to a: the point $x = a$ would force $y \leq 0$.
- Using $N = \frac{1}{2}(2a - y)$ for odd y, which isn't an integer.
- Trying to find $\mathrm{E}(Y)$ from the two-case marginal. It works, but summing over columns of the joint distribution avoids the parity split altogether.

Beyond STEP

Getting a marginal from a joint distribution means summing out the other variable, and the hard part is always the limits of the region. With continuous variables the same task becomes choosing the limits in a double integral, and swapping the order of summation (as we did for $\mathrm{E}(Y)$) becomes swapping the order of integration. Grimmett and Welsh, *Probability: An Introduction*, covers joint and marginal distributions for both discrete and continuous variables.

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